

George Padilla

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Education

Cornell University (GPA 4.1/4.3)

Ithaca, NY

Ph.D. in Materials Science and Engineering

On-going

- NSF GRFP Fellow

Cornell University (GPA 4.1/4.3)

Ithaca, NY

M.S. in Materials Science and Engineering

08/2025

- ACS Bridge Fellow

Temple University (GPA 3.7/4.0)

Philadelphia, PA

B.S. in Biology

05/2022

Technical Skills

Programming: Python, Bash, C++, Java, R, MATLAB, SLURM

Simulations: Molecular Dynamics, Monte Carlo (HOOMD-blue and LAMMPS), Density Functional Theory (GAMESS)

Analysis & Computation: Machine Learning (Unsupervised and Supervised) Techniques, Graph Neural Networks, Voronoi Tessellation, Percolation Theory, Gaussian Mixture Models, Dynamic Programming, Algorithm Development, Academic Writing

Research Experience

Cornell University, Dept. of Materials Science & Engineering

Ithaca, NY

Graduate Researcher, CAPE CryStAL

09/2023 – *Present*

- **Project:** Computational Modeling of Active Fluids
- Developed novel models to simulate activity through particle interconversion within Lennard-Jones binary fluids using molecular dynamics simulations and Monte Carlo methods.
- Derived phase order parameters through unsupervised ML clustering of simulation datasets.

University of Pennsylvania, Dept. of Genetics

Philadelphia, PA

Research Assistant II, Shalem Lab

09/2022 – 07/2023

- **Project:** Arrayed CRISPR-KO Screen in iPSC-derived Neurons to Identify Novel Modifiers of ALS Pathology
- Extracted data from confocal imaging of neurons using FIJI and determined TDP-43 aggregation levels using R.

University of Pennsylvania, Dept. of Biochemistry & Biophysics

Philadelphia, PA

Undergraduate Researcher, Shorter Lab

05/2021 – 07/2022

- **Project:** In-Vitro Analysis of RNA Binding on TDP-43 Liquid-Liquid Phase Separation
- Employed fast protein liquid chromatography for protein purification and assayed various TDP-43 + RNA conditions for TDP-43 aggregation with spectroscopy and microscopy.

Integral Molecular

Philadelphia, PA

Research Intern, Membrane Proteome Array Team

07/2021 – 09/2021

- Screened for off-target antibody binding using Integral Molecular's membrane proteome array platform.

Wistar Institute

Philadelphia, PA

Biomedical Technician Training (BTT) Program

07/2021 – 09/2021

- Completed intensive training in biomedical techniques.

Awards & Scholarships

2025: Engagement Award, Greater Ithaca Activity Center (GIAC).
2025: MSE Distinguished Awards Gala 2nd Place Poster Award, Cornell University.
2024: NSF Graduate Research Fellowship Program Award, Cornell University.
2024: ACS Fall 2024 Bridge Travel Grant, Cornell University.
2023: ACS Ezra's Bridge Fellowship, Cornell University.
2022: Isadore Schrager CCP Scholarship, Temple University.
2021: Broad Street Finish Line Scholarship, Temple University.

Publications

Padilla, G., Dufresne, E., Dshemuchadse, J. (2025–).
Emergent spinodal architectures in active fluids of interconverting particles.
In preparation.

Copley, K. E., Rubien, J. D., Smirnov, A., **Padilla, G.**, Shorter, J. (2026).
Rational Design of RNA Chaperones to Antagonize TDP-43 Condensation.
ACS Chemical Biology, in submission.

Guo, L., Mann, J. R., Mauna, J. C., Copley, K. E., Wang, H., Rubien, J. D., Odeh, H. M., Lin, J., Lee, B. L., Ganser, L., Robinson, E., Kim, K. M., Murthy, A. C., Paul, T., Portz, B., Gleixner, A. M., Diaz, Z., Carey, J. L., Smirnov, A., **Padilla, G.**, Lavorando, E., Espy, C., Shang, Y., Huang, E. J., Chesi, A., Fawzi, N. L., Myong, S., Shorter, J. (2026).
Defining RNA oligonucleotides that reverse deleterious phase transitions of RNA-binding proteins with prion-like domains.
Molecular Cell. doi:10.1016/j.molcel.2025.12.009

Presentations

05/2025: Cornell University Materials Science and Engineering Distinguished Awards Gala – “Modeling Liquid-Liquid Phase Separation with Interconversion-Driven Size-Asymmetric Active Fluids”.

04/2022: American Society for Biochemistry and Molecular Biology Annual Meeting – “Exploring the Effect of RNA Binding on TDP-43 Liquid-Liquid Phase Separation”.

12/2021: UPenn Biochemistry & Biophysics Annual Retreat – “The Effect of RNA Binding on TDP-43 Liquid-Liquid Phase Separation”.

11/2021: Temple CST Undergraduate Research Symposium – “The Effect of RNA Binding on TDP-43 Liquid-Liquid Phase Separation”.

Activities & Leadership

Bright Minds Science Hour, GIAC & Cornell University	Ithaca, NY
President/Instructor	01/2024 – Present
• Launched a monthly STEM outreach program offering hands-on experiments and interactive science discussions for young learners.	
JEDI Group, Cornell University	Ithaca, NY
Member/Presenter/Organizer	09/2023 – Present
• Collaborated with fellow MSE graduate students and faculty to design and implement community-building initiatives within MSE, including workshops and panel discussions.	
Penn Medicine Hospice Services, University of Pennsylvania	Philadelphia, PA
Hospice Volunteer	2021 – 2023
Gray Matters, Temple University	Philadelphia, PA
Scientific Editor	2022
Health Link Society, Temple University	Philadelphia, PA
Member/Volunteer	2021 – 2022

Undergraduate Research Program, Temple University

Researcher

Philadelphia, PA

2021 – 2022

Louis Stokes Alliances for Minority Participation, Temple University

Member

Philadelphia, PA

2020 – 2021

Lighthouse Inc.

Summer Program Counselor

Philadelphia, PA

2018 – 2019